

Classifying Cancer Types from Gene Expression Profiles Using Supervised Learning

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Abstract—Gene expression profiles have long been a compelling resource for classifying cancer types. However, many traditional statistical methods cannot scale to handle extremely high-dimensional datasets such as those used in genomics. Rather than relying on these traditional methods, this study leverages three supervised machine learning models (logistic regression, random forest, and XGBoost) to classify cancer samples from The Cancer Genome Atlas (TCGA) into cancer types based on their gene expression profiles. High dimensional gene expression profiles are first reduced to 100 principal components via principal component analysis (PCA), and group-aware cross-validation is used to ensure samples from the same patient are grouped in the same folds during training. Each classifier yielded accuracy and F1-scores above 0.90, indicating promising predictive capabilities for all three models. Although these current models assume that sample inputs are cancerous, future work could focus on extending them to datasets that include non-cancerous samples.

I. INTRODUCTION

Gene expression can provide key insights to cell type, function, and health. Cells in different tissues often express vastly different genes, allowing them to specialize in their functionality. In cancerous cells, however, gene expression is atypical for a cell of its type. High expression of a gene in one cell type can be normal whereas the same level of expression can be indicative of cancer in another cell type. These gene expression anomalies can therefore be used to classify cells into different cancer types. This study will explore several multi-class supervised machine learning models that use gene expression profiles to predict cancer type.

II. RELATED WORK

Leveraging machine learning to classify cell samples into cancer type based on their gene expression is not a novel idea, as this has been an active field of research for decades. For example, in their paper titled “Ensemble machine learning on gene expression data for cancer classification”, Aik Choon Tan and David Gilbert developed three supervised machine learning models (namely, a C4.5 decision tree, a bagged decision tree, and a boosted decision tree) to classify tumors based on gene expression profiles. Their study indicated that ensemble supervised methods tend to outperform single models [1]. This is consistent with most applications of ensemble methods; ensemble methods tend to generalize better than single models and are less susceptible to overfitting. Given the success of ensemble methods in Tan and Gilbert’s study, two of the three

models in this study will also be ensemble methods (namely, Random Forest and XGBoost).

III. METHODOLOGY

A. Data Collection

The datasets used by the models were both provided by The Cancer Genome Atlas (TCGA). The first dataset, “gene expression RNAseq - Batch effects normalized mRNA data”, contains the gene expression profiles for various tissue samples [2]. In this dataset, rows represent genes, columns represent samples, and table values represent gene expression as a floating point value, measured in $\log_2(norm_value + 1)$. In order to properly join this table to the second dataset, the gene expression dataset was transposed such that rows represent samples and columns represent genes. Outliers in this dataset were identified but kept, as outliers in genomic datasets often hold legitimate biological significance.

The second dataset, titled “phenotype - Curated clinical data”, contains patient survival data [3]. In this dataset, rows represent samples, columns represent clinical metrics, and table values are measurements of various data types. Although many of the features in this dataset are not relevant to this study, this dataset contains a feature titled “cancer type abbreviation”, which includes three to four letter string codes representing the cancer diagnosis of the patient [4]. These nominal categorical labels represent the predicted output of the models.

Prior to conducting data pre-processing and extensive exploratory data analysis (EDA), careful consideration of the aforementioned datasets was required. Importantly, the cancer diagnosis label is prescribed to the patient, not the sample. Further, a single patient may have multiple samples. Lastly, samples can be from various tissue types. To account for these nuances, information was extracted from the TCGA’s standardized sample names, which are of the form “TCGA-site code-patient code-sample type code”. As explained in the TCGA’s documentation, a sample type code of “01” represents a primary tumor (ie, that sample is responsible for the diagnosis of the patient) [5]. Therefore, all samples not of sample type “01” were dropped from the dataset. Patient codes were extracted from sample names in order to group samples by patient during model development. The two initial datasets

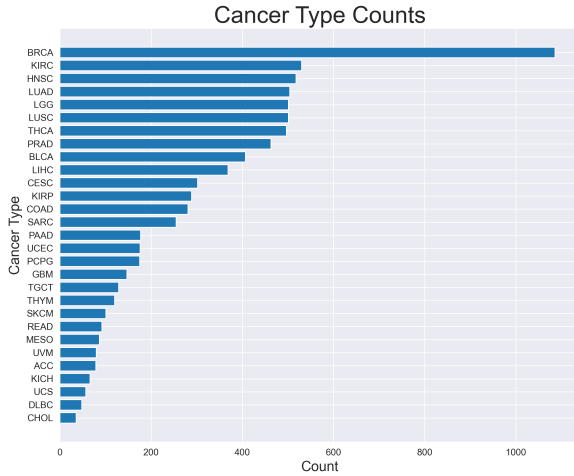


Fig. 1. Output Class Imbalance

were then joined based on their sample names to create a single dataframe for model development.

B. Data Pre-processing and Exploratory Data Analysis

The previous steps resulted in a dataset with 8,080 rows and 20,534 features. Before conducting feature selection, class imbalance was detected using a simple bar chart, shown in Figure 1. Fig. 1 indicates that the dataset suffered from extreme class imbalance, as 1,086 records were labeled Breast invasive carcinoma (BRCA), and some cancer types such as Ovarian serous cystadenocarcinoma (OV), Stomach adenocarcinoma (STAD), and Esophageal carcinoma (ESCA) had less than five samples. Cancer types with fewer than 5 samples were dropped, as there was insufficient data to develop statistically significant results. Remaining class imbalance was largely handled with Synthetic Minority Oversampling Technique (SMOTE) and setting the 'class_weight' parameter to 'balanced' for applicable models.

Given the sheer size of the dataset, many traditional feature selection and EDA methods were not feasible. Instead, principal component analysis (PCA) was used to drastically reduce the dimensionality of the input data while still preserving an acceptable amount of variance from the initial dataset. PCA also addresses multicollinearity concerns, as principal components are, by definition, orthogonal and linearly-independent. Note that, in order to implement PCA, the data must first be centered and scaled. Therefore, all input data was transformed with scikit-learn's StandardScaler object.

Importantly, PCA was applied on the training dataset only. Fitting the PCA model to the entire dataset would result in data leakage and yield misleading results. Therefore, training and testing datasets were created prior to PCA. These datasets were created with scikit-learn's GroupShuffleSplit object to prevent additional data leakage by ensuring samples from the same patient were grouped together when splitting the data.

The first 250 principal components of the training dataset were computed, and their percent explained variance was

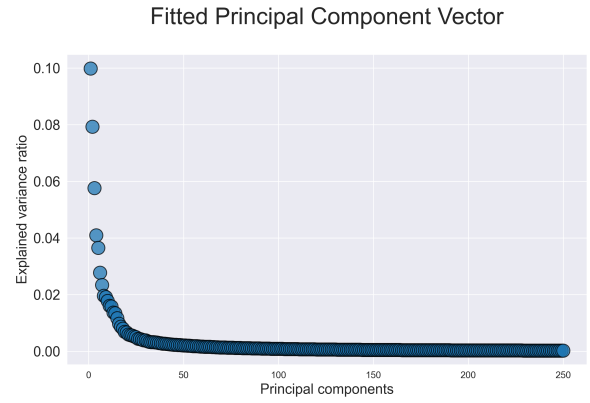


Fig. 2. Variance by Principal Components

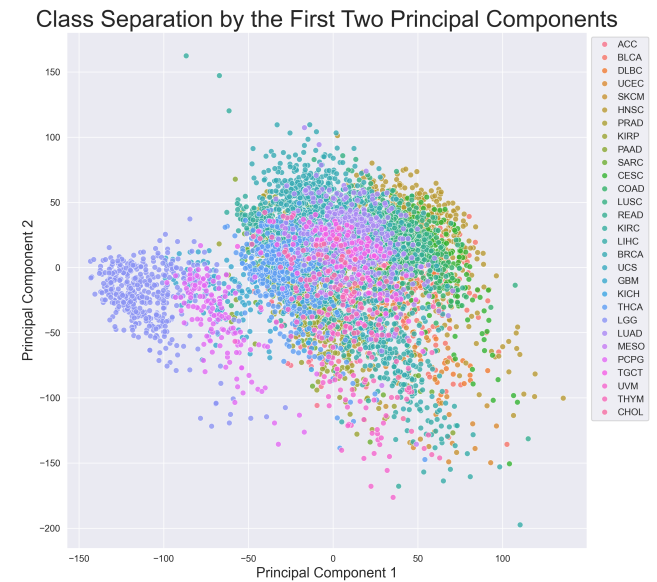


Fig. 3. Class Separation by Principal Components

plotted as shown in Fig. 2. Because SMOTE scales poorly with input dimensionality, fewer principal components led to better performance. However, simply relying on the elbow point of the graph in Fig. 2 results in 30 principal components which only explain about 58% of the original dataset's variance. To balance performance and explained variance, the first 100 principal components were used, which explained approximately 70% of the initial variance.

In dropping statistically insignificant cancer types and transforming the dataset with PCA, the dimensionality dropped from 8,080 samples to 8,073 samples, and from 20,531 features to 100 features. A scatterplot of the first two principal components was used to visualize initial class separation, as shown in Fig. 3. Although class separation is minimal with only two principal components, some classes already exhibit significant separation from the others. Most notably, Brain

Lower Grade Glioma (LGG) displays substantial separation from other classes along the first principal component, indicating that partial class separation has already been achieved with only two principal components.

C. Model Development

Three different classifiers (Logistic Regression, Random Forest, and XGBoost) were chosen to predict cancer type from the gene expression profiles. These classifiers were chosen due to their ability to handle high-dimensional data and multiclass classification. For each classifier, a base model was trained without cross-validation, a second model was trained with cross-validation, and a final model was created with cross-validation and hyperparameter tuning, resulting in a total of nine models. The following paragraphs will explain the methodology used in the training of the base models, cross-validated models, and tuned models, respectively.

For each base model, the entire training dataset was transformed via StandardScaler and PCA with 100 components. The transformed data was then oversampled with SMOTE and the model was trained using the transformed, oversampled data. Default parameters (excluding random_state=42 for reproducibility and n_jobs=-1 to make use of all available CPUs) were used for all base models. For XGBoost, the output variable (y_train) was label encoded, as XGBoost requires integer output classes. It is worth noting that the base models are largely uninformative, as they do not leverage cross-validation. This can result in overfit models and overly-optimistic results.

Cross-validation was achieved in a group-aware manner via 3 group-aware folds. As with the original group-aware train-test split, transformations and oversampling must occur on the fold-level to prevent data leakage. In order to achieve this behavior, imblearn’s Pipeline object was used in the training of each model. The pipelines ensure that scaling, PCA, and SMOTE are applied only to the training folds in each step of cross-validation. As with the base models, default parameters were used when training the cross-validated models.

The three final models underwent cross-validation as described above as well as hyperparameter tuning. Tuning was achieved with grid search on the hyperparameters of the classifier in each pipeline. For the logistic regression classifier, only the C and penalty parameters were be tuned. Both L1 and L2 regularization were tested as penalties, and C values of 0.1, 1, and 10 were tested to control the strength of the regularization term. The solver was held constant as ‘saga’, as this solver supports multinomial classification as well as both L1 and L2 regularization. The class_weight parameter was also held constant to account for class imbalance. Lastly, max_iter was held constant at 250. Although the relatively low max_iter value resulted in failure to converge warnings, higher values also failed to converge and significantly increased training times. The failure to converge is likely due to the high dimensionality of the input feature space.

For the final random forest classifier, the n_estimators, max_depth, and min_samples_leaf parameters were tuned.

TABLE I
ACCURACY OF EACH MODEL

Classifier	Base Model	Cross-Validated Model	Tuned Model
Logistic Regression	0.953	0.949	0.951
Random Forest	0.944	0.936	0.937
XGBoost	0.935	0.929	0.936

TABLE II
F1-SCORES OF FINAL MODELS

Classifier	Macro Average	Weighted Average
Logistic Regression	0.92	0.96
Random Forest	0.91	0.94
XGBoost	0.92	0.95

As with logistic regression, the class_weight parameter is held constant to account for class imbalance. Lastly, the n_estimators, max_depth, and learning_rate parameters were tuned for the final XGBoost classifier. The objective and num_class parameters were held constant to ‘multi:softmax’ and the number of output classes, respectively. This enabled XGBoost to perform multiclass classification, and num_class simply specified the number of classes.

IV. RESULTS AND DISCUSSION

The performance of each model was initially measured simply by their accuracies and models were compared per classifier. Each of these accuracy scores is recorded in Table I. All three classifiers received slightly lower accuracies when cross-validated when compared to their base models. These lower accuracy scores are to be expected; the base models are likely overfit due to the lack of cross-validation and the resulting scores are overly-optimistic. As expected, the accuracy of each classifier increased after incorporating hyperparameter tuning. However, these improvements are marginal, indicating that the default hyperparameters for each classifier are nearly optimal. All models, including base models and un-tuned models, yielded accuracy scores greater than 0.90, suggesting that the transformed data is well separated.

Final classifiers were compared with their normalized confusion matrices (see Fig. 4) and classification reports. Overall, each classifier had very few erroneous classifications. Across all models, colon adenocarcinoma (COAD) and rectum adenocarcinoma (READ) were the most commonly confused classes. Random Forest, for example, erroneously classified 80% of COAD cases as READ. These misclassifications are likely due to the many similarities of these two cancer types. As stated by the Cancer Genome Atlas, “non-hypermuted adenocarcinomas of the colon and rectum are not distinguishable at the genomic level” [6]. These poor results of all three classifiers, supplemented with existing literature, suggest that certain types of cancer cannot be differentiated from gene expression alone.

The classification reports further reinforced the conclusions drawn from the confusion matrices. F1-scores were primarily used when evaluating model correctness, as F1-scores are less sensitive to class imbalance than raw accuracy. As shown by

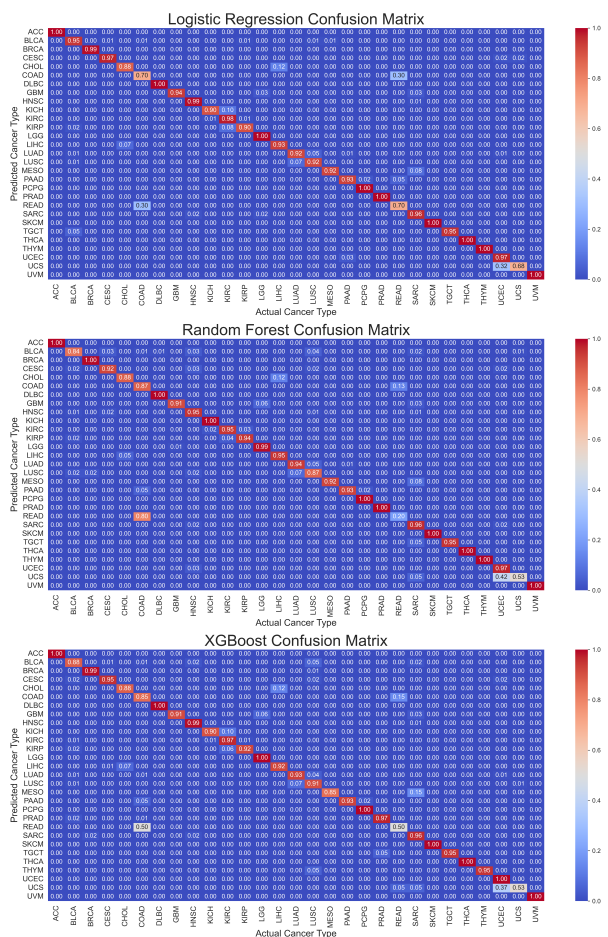


Fig. 4. Confusion Matrices of Final Classifiers

Table II, weighted and macro F1-scores of all three classifiers were above 0.90, indicating few misclassifications. Inspection of the F1-scores per class reveals low scores for READ and COAD across all models, further demonstrating that the classifiers struggle to correctly discern these two cancer types.

V. CONCLUSION

This study explored using various supervised learning techniques to classify cancer type based on gene expression. All models achieved accuracy above 0.90, indicating high predictive capabilities. Careful analysis of results, however, revealed that all final models performed poorly when differentiating specific cancer types, especially COAD and READ. This can be explained by the genetic similarities of certain cancer types. Further, despite implementing SMOTE oversampling and setting `class_weight='balanced'` when applicable, the severe class imbalance likely influenced results. In order to develop a more general and holistic model, a more balanced dataset can be used.

One of the greatest limitations of the models in this study is that they assume that input samples are cancerous. Although the original TCGA datasets do include non-cancerous samples, these samples were deliberately excluded in model develop-

ment. Recall that the datasets provided by TCGA assign the cancer diagnosis to the patient and not the sample, and that the tissue type of each sample is indicated by the sample barcode. However, the non-cancer tissue types are far too broad to draw meaningful biological insights. Future improvements on these models could be largely focused on accounting for non-cancerous samples, as this would eliminate the assumption that sample types are cancerous and lead to a more insightful model.

In summary, logistic regression, random forest, and XGBoost all demonstrated high predictive abilities when classifying tissue samples based on gene expression. Given their similar metrics, all models performed roughly the same when predicting cancer type and no classifier is necessarily preferable to the others. Despite the promising metrics for each of these models, genetic similarities between certain cancer types and insufficient data on minority classes significantly limits the use cases of these models. Rather than relying on these models, traditional methods such as bioimaging and histological analysis should be used when diagnosing cancer. Nonetheless, this study offers insights as to how supervised learning techniques can be leveraged to classify cells based on high-dimensional gene expression data.

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